The Collected Mathematical Papers of Arthur Cayley, Vol. III

PDF Pages 288–300 (printed pp. 266–278)

Papers 210, 211, 212 (Paper 212 begins; continues beyond p. 300)

210.

ON THE CUBIC TRANSFORMATION OF AN ELLIPTIC FUNCTION.

[From the *Philosophical Magazine*, vol. xv. (1858), pp. 363–364.]

Let

$$z = \frac{(a',\, b',\, c',\, d' \mid x, 1)^3}{(a,\, b,\, c,\, d \mid x, 1)^3}$$

be any cubic fraction whatever of x, then it is always possible to find quartic functions of z, x respectively, such that

$$\frac{dz}{\sqrt{(\alpha,\, \beta,\, \gamma,\, \delta,\, \epsilon \mid z, 1)^4}} = \frac{dx}{\sqrt{(A,\, B,\, C,\, D,\, E \mid x, 1)^4}}.$$

This depends upon the following theorem, viz. putting for shortness,

$$U = (a,\, b,\, c,\, d \mid x,\, y)^3, \quad U' = (a',\, b',\, c',\, d' \mid x,\, y)^3,$$

and representing by the notation

$$\text{disct.. } (aU' - a'U,\ bU' - b'U,\ cU' - c'U,\ dU' - d'U);$$

or more shortly by

$$\text{disct.. } (aU' - a'U, \ldots),$$

the discriminant in regard to the facients (λ, μ) of the cubic function

$$(aU' - a'U,\ bU' - b'U,\ cU' - c'U,\ dU' - d'U \mid \lambda,\, \mu)^3;$$

or what is the same thing, the cubic function

$$\begin{gathered}(a,\, b,\, c,\, d \mid \lambda,\, \mu)^3 \cdot (a',\, b',\, c',\, d' \mid x,\, y)^3 \\ -\, (a',\, b',\, c',\, d' \mid \lambda,\, \mu)^3 \cdot (a,\, b,\, c,\, d \mid x,\, y)^3;\end{gathered}$$

and by $J(U, U')$ the functional determinant, or Jacobian, of the two cubics U, U'; the theorem is that the discriminant contains as a factor the square of the Jacobian, or that we have

$$\text{disct.. } (aU' - a'U, \ldots) = \{J(U, U')\}^2 \cdot (A,\, B,\, C,\, D,\, E \mid x,\, y)^4.$$

For assuming this to be the case, then (disregarding a mere numerical factor) we have

$$U\, dU' - U'\, dU = J(U, U')(y\, dx - x\, dy),$$

and the two equations give

$$\frac{U\, dU' - U'\, dU}{\sqrt{\text{disct.. } (aU' - a'U, \ldots)}} = \frac{y\, dx - x\, dy}{\sqrt{(A,\, B,\, C,\, D,\, E \mid x,\, y)^4}},$$

whence writing z for $U' \div U$, and putting y equal to unity, we have

$$\frac{dz}{\sqrt{\text{disct.. } (az' - a'z, \ldots)}} = \frac{dx}{\sqrt{(A,\, B,\, C,\, D,\, E \mid x, 1)^4}},$$

where disct.. $(az' - a'z, \ldots)$, or at full length,

$$\text{disct.. } (az' - a',\ bz' - b',\ cz' - c',\ dz' - d'),$$

is a given quartic function of z,

$$= (\alpha,\, \beta,\, \gamma,\, \delta,\, \epsilon \mid z, 1)^4$$

suppose; and this proves the theorem of transformation.

The assumed subsidiary theorem may be thus proved: suppose that the parameter θ is determined so that the cubic

$$U + \theta U'$$

may have a square factor, the cubic may be written

$$(a + \theta a',\ b + \theta b',\ c + \theta c',\ d + \theta d' \mid x,\, y)^3,$$

and the requisite condition is

$$\text{disct.. } (a + \theta a', \ldots) = 0;$$

there are consequently four roots; and calling these θ_1, θ_2, θ_3, θ_4, we have identically

$$\text{disct.. } (a + \theta a', \ldots) = K(\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

or what is the same thing,

$$\text{disct.. } (aU' - a'U, \ldots) = K(U + \theta_1 U')(U + \theta_2 U')(U + \theta_3 U')(U + \theta_4 U').$$

Now any double factor of U or U' (that is the linear factor which enters twice into U or U') is a simple factor of $J(U, U')$, and we have $J(U, U') = J(U,\, U + \theta U')$, and consequently

$$J(U, U') = J(U',\, U + \theta U') = \&c.;$$

hence the double factors of each of the expressions $U + \theta_1 U'$, $U + \theta_2 U'$, $U + \theta_3 U'$, $U + \theta_4 U'$ are simple factors of $J(U, U')$, or what is the same thing, $J(U, U')$ is the product of four linear factors, which are respectively double factors of the product

$$(U + \theta_1 U')(U + \theta_2 U')(U + \theta_3 U')(U + \theta_4 U'),$$

or this product contains the factor $\{J(U, U')\}^2$, which proves the theorem.

2, Stone Buildings, W.C., March 5, 1858.

211.

ON A THEOREM RELATING TO HYPERGEOMETRIC SERIES.

[From the *Philosophical Magazine*, vol. xvi. (1858), pp. 356–357.]

In attempting to verify a formula of Hansen's relating to the development of the disturbing function in the planetary theory, I was led to a theorem in hypergeometric series: viz. writing, as usual,

$$F(\alpha, \beta, \gamma, x) = 1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma} x + \frac{\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1}{1 \cdot 2 \cdot \gamma \cdot \gamma + 1} x^2 + \cdots$$

then the product

$$F(\alpha, \beta, \gamma + \tfrac{1}{2}, x)\, F(\gamma - \alpha, \gamma - \beta, \gamma + \tfrac{1}{2}, x)$$

is connected with

$$(1 - x)^{\gamma - \alpha - \beta}\, F(2\alpha, 2\beta, 2\gamma, x)$$

by a simple relation: for if the last-mentioned expression is put equal to

$$1 + Bx + Cx^2 + Dx^3 + \cdots$$

then the product in question is equal to

$$1+\frac{\gamma}{\gamma+\frac{1}{2}}Bx+\frac{\gamma\cdot\gamma+1}{\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}}Cx^2+\frac{\gamma\cdot\gamma+1\cdot\gamma+2}{\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}}Dx^3+\&c.$$

The form of the identity thus arrived at will be best perceived by considering a particular case. Thus, comparing the coefficients of x^3, we have

$$\left.\begin{array}{l}\dfrac{\alpha\cdot\alpha+1\cdot\alpha+2\cdot\beta\cdot\beta+1\cdot\beta+2}{1\cdot2\cdot3\cdot\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}}\cdot1\\ +\dfrac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot2\cdot\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}}\cdot\dfrac{\gamma-\alpha\cdot\gamma-\beta}{1\cdot\gamma+\frac{1}{2}}\\ +\dfrac{\alpha\cdot\beta}{1\cdot2}\cdot\dfrac{\gamma-\alpha\cdot\gamma-\alpha+1\cdot\gamma-\beta\cdot\gamma-\beta+1}{1\cdot2\cdot\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}}\\ +1\cdot\dfrac{\gamma-\alpha\cdot\gamma-\alpha+1\cdot\gamma-\alpha+2\cdot\gamma-\beta\cdot\gamma-\beta+1\cdot\gamma-\beta+2}{1\cdot2\cdot3\cdot\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}}\end{array}\right\}=\frac{\gamma\cdot\gamma+1\cdot\gamma+2}{\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}}D$$

It may be observed that the function on the right-hand side is, as regards α, a rational and integral function of the degree 3, and as such may be expanded in the form

$$\left.\begin{array}{l}A\,\alpha\cdot\alpha+1\cdot\alpha+2\\ +B\,\alpha\cdot\alpha+1\\ +C\,\alpha\\ +D\end{array}\right\}\begin{array}{l}\cdot-\gamma-\alpha,\\ \cdot-\gamma-\alpha\cdot\gamma-\alpha+1,\\ \cdot-\gamma-\alpha\cdot\gamma-\alpha+1\cdot\gamma-\alpha+2,\end{array}$$

and that the last coefficient D can be obtained at once by writing $\alpha=0$; this in fact gives

$$D\gamma\cdot\gamma+1\cdot\gamma+2=\frac{\gamma-\beta\cdot\gamma-\beta+1\cdot\gamma-\beta+2}{1\cdot2\cdot3}\cdot\frac{\gamma\cdot\gamma+1\cdot\gamma+2}{\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}},$$

and thence

$$D=\frac{\gamma-\beta\cdot\gamma-\beta+1\cdot\gamma-\beta+2}{1\cdot2\cdot3\cdot\gamma+\frac{1}{2}\cdot\gamma+\frac{3}{2}\cdot\gamma+\frac{5}{2}},$$

which agrees with the left-hand side of the equation; and the value of the first coefficient A may be obtained in like manner with a little more difficulty; but I have not succeeded in obtaining a direct proof of the equation. The form of the equation shows that the left-hand side should vanish for $\gamma=-\frac{1}{2}$, which may be at once verified.

Grassmere, August 25, 1858.

212.

A MEMOIR ON THE PROBLEM OF DISTURBED ELLIPTIC MOTION.

[From the *Memoirs of the Royal Astronomical Society*, vol. xxvii., 1859, pp. 1–29.
Read March 9, 1858.]

I venture to take up the problem of disturbed elliptic motion, for the sake of a further elaboration of the analytical theory. The points which present difficulty are the measurement of longitudes in the varying plane of the orbit, and (in the lunar theory) the determination of the position of the orbit by reference to the varying plane of the sun's orbit; it is, in memoirs and works on the lunar and planetary theories, often difficult to discover where or how (or whether at all) account is taken of these variations, and the analytical mode of treatment is for the most part very imperfect. I must except always Hansen's *Fundamenta Nova* [*investigationis orbitæ veræ quam Luna perlustrat*, Gotha 1838] where the points referred to are treated in a perfectly rigorous manner. There is, however, a want of clearness in the form under which his investigations are presented; and the comprehension of them is greatly facilitated by Jacobi's remarks, published under the title "*Auszug zweier Schreiben des Prof. Jacobi an Herrn Director Hansen*" (*Crelle*, t. XLII. pp. 12–31

(1851)). Jacobi observes that the integration of Hansen's system of differential equations introduces seven arbitrary constants, which, in the expressions for the coordinates referred to fixed axes, reduce themselves to six. The seventh constant, neglecting the disturbing forces, is in fact a constant which determines the position in the orbit of the arbitrary origin from which the longitudes in orbit are reckoned. I have, in my paper "On Hansen's Lunar Theory," *Quarterly Mathematical Journal*, vol. I. pp. 112–125 (1855), [163], termed this origin the "departure-point," and longitudes measured from it "departures." The seventh constant may be taken to be the departure of the node. I reproduce in the present memoir the explanation of what is meant by the departure when the plane of the orbit is variable. If the problem is treated by the method of the variation of the elements, the seventh constant becomes, like the other elements, variable; and we have thus a seventh variable element, the departure of the node. The element just referred to (the departure of the node) forms, with the longitude of the node and the inclination, a group of three elements, which determine the position of the orbit and of the departure-point. The coordinates of the planet are in the first instance taken to be the radius vector, longitude, and latitude; but the before-mentioned three elements being considered as given, the position of the planet depends only on the radius vector and the departure. These may be then expressed in terms of the remaining four elements; as to the choice of these four elements, it is to be remarked that there is one element which only enters through the mean anomaly, and that there is great convenience in representing with Hansen the mean anomaly by a single letter; and that in the various formulæ we may use, in the place of the element implicitly involved in the mean anomaly, the mean anomaly itself, or treat the mean anomaly as an element; the four elements may be taken to be the semi-axis major, the eccentricity, the mean anomaly, and the departure of the pericentre. And joining to these the before-mentioned three elements, we have the system of elements represented in the memoir by a, e, g, ϖ, θ, ϕ. It has been assumed so far that the three elements determine the position of the orbit and departure-point in reference to a fixed plane and origin of longitudes; but we may suppose more generally that, instead of the fixed plane and origin of longitudes, we have a variable plane or orbit of reference and a departure-point in this variable orbit of reference. The quantities which determine the orbit of reference and departure-point are naturally taken to be the departure of the node, longitude of the node, and inclination; these are assumed to be given functions of the time, and they are in the memoir represented by σ', θ', ϕ'. The three elements of the planet's orbit (viz. departure of node, longitude of node, and inclination) in relation to the orbit of reference and departure-point therein, are in the memoir represented by Σ, Θ, Φ, and the system of elements ultimately adopted is therefore a, e, g, ϖ, Σ, Θ, Φ. I obtain formulæ for the variations of these elements under two different modes of expression of the disturbing function: first, when the disturbing function is expressed in terms of the radius vector and departure and of the three elements Σ, Θ, Φ; secondly, when the disturbing function is expressed in terms of the seven elements a, e, g, ϖ, Σ, Θ, Φ. The establishment of the two sets of formulæ just referred to constitutes the chief object of the memoir; but the memoir contains some other investigations and formulæ in relation to the general subject.

The coordinates of the planet are

r, the radius vector,
v, the longitude,
y, the latitude.

The attractive force at distance unity is for convenience represented by n^2a^3, which denotes, therefore, an absolute constant; but the significations of n and a are not yet defined.

The disturbing function, as used by Lagrange, is denoted by Ω, that is $\Omega = -R$, if R be the disturbing function of the *Mécanique Céleste.*

The equations of motion are

$$\frac{d}{dt}\frac{dr}{dt} - r\cos^2 y\left(\frac{dv}{dt}\right)^2 - r\left(\frac{dy}{dt}\right)^2 + \frac{n^2a^3}{r^2} = \frac{d\Omega}{dr},$$
$$\frac{d}{dt}\left(r^2\cos^2 y\,\frac{dv}{dt}\right) = \frac{d\Omega}{dv},$$
$$\frac{d}{dt}\left(r^2\frac{dy}{dt}\right) + r^2\cos y\sin y\left(\frac{dv}{dt}\right)^2 = \frac{d\Omega}{dy},$$

where Ω is regarded as a function of r, v, y, or (as this may be expressed) where $\Omega = \Omega(r, v, y)$.

If we neglect the disturbing forces, the planet moves in an ellipse; and taking a to represent the semi-axis major, the mean motion will be n. The mean anomaly, which I call g, will be a function of the form $nt + c$; but as c only enters through g, it will be convenient to use the mean anomaly g (considered as implicitly involving an arbitrary constant c) in the place of an element, and I write

a, the semi-axis major,
e, the eccentricity,
g, the mean anomaly,
θ, the longitude of node,
ϕ, the inclination,
$\mathfrak{C}$, the distance of pericentre from node.

I assume also

f, the true anomaly,
z, the distance of planet from node,
x, the reduced distance from node.

We have then r and f given functions of t and the elements, viz. we may write

$$r = a\,\text{elqr.}(e, g), \qquad f = \text{elta.}(e, g),$$

(read elqr. elliptic quotient radius, and elta. elliptic anomaly). These values satisfy $r = \dfrac{a(1-e^2)}{1+e\cos f}$. Moreover z, x, y, are the hypothenuse, base, and perpendicular of a right-angled spherical triangle, the base angle whereof is ϕ; the equations which connect these quantities are therefore

$$\begin{aligned} \tan x &= \tan z \cos\phi, \\ \sin y &= \sin z \sin\phi, \\ \tan y &= \sin x \tan\phi, \\ \cos z &= \cos x \cos y, \end{aligned}$$

equivalent, of course, to two equations. The first and second of them give in fact x, y, in terms of z and ϕ. The value of z is

$$z = \mathfrak{C} + f,$$

so that x and y are given functions, and the longitude v is given in terms of x and θ by the equation

$$v = x + \theta,$$

and consequently the three coordinates r, v, y, are by the system of equations given in terms of t and the elements.

From the equations which connect z, x, y, ϕ, treating all these quantities as variable we deduce

$$\begin{aligned} \sec^2 x\,dx &= \cos\phi\sec^2 z\,dz - \tan z\sin\phi\,d\phi, \\ \cos y\,dy &= \sin\phi\cos z\,dz + \sin z\cos\phi\,d\phi, \\ \sec^2 y\,dy &= \tan\phi\cos x\,dx + \sin x\sec^2\phi\,d\phi, \\ \sin z\,dz &= \cos y\sin x\,dx + \cos x\sin y\,dy, \end{aligned}$$

equivalent of course to two equations; and the system is easily reduced to the more convenient form

$$\begin{aligned} dx &= \cos\phi\sec^2 y\,dz - \tan z\cos^2 x\sin\phi\,d\phi, \\ dy &= \sin\phi\cos x\,dz + \cos x\sin x\cos\phi\,d\phi, \\ dx &= \cos\phi\sec^2 y\,dy - \tan z\cos\phi\cot x\cos\phi\,d\phi, \\ dz &= \cos\phi\,dy + dx\ \cos x\sin\phi\,d\phi, \end{aligned}$$

joining to these equations the

$$dz = d\mathfrak{C} + df, \qquad dv = dx + d\theta,$$

and considering at present the mere analytical forms, first if $d\phi = 0$, $d\mathfrak{C} = 0$, we have

$$\begin{aligned} dx &= \cos\phi \sec^2 y \, dz, \\ dy &= \sin\phi \cos x \, dz, \\ dz &= df, \\ dv &= dx. \end{aligned}$$

Next, if $dy = 0$, $dv = 0$, we have

$$\begin{aligned} dx &= -\tan z \operatorname{cosec}\phi \, d\phi, \\ dz &= -\tan z \cot\phi \, d\phi, \\ d\mathfrak{C} &= \cos\phi \, dx, \\ dx &= d\theta, \\ dz + \cos\phi \, d\theta &= 0. \end{aligned}$$

I remark also that the equation $\tan y = \sin x \tan\phi$ may be written in the form

$$\cos^2\phi + \sec^2 y \sin^2\phi \cos^2 x = 1.$$

The equations

$$r = a \, \mathrm{elqr.}(e, g), \qquad f = \mathrm{elta.}(e, g),$$

treating all the quantities as variable, give

$$dr = \frac{ae\sin f}{\sqrt{1-e^2}} \, dg + \frac{1-e^2}{1+e\cos f} \, da - a\cos f \, de,$$

$$df = \frac{(1+e\cos f)^2}{(1-e^2)^{3/2}} \, dg + \frac{\sin f(2+e\cos f)}{1-e^2} \, de,$$

to which is to be joined

$$\frac{d^2r}{dt^2} = ae(1-e^2)\sin f \, df + \frac{1-e^2}{(1+e\cos f)^2} \, da + \frac{a(2e-1+e^2\cos f)}{(1+e\cos f)^2} \, de,$$

all which formulæ will be useful.

If we treat the elements as constant, then in the foregoing expressions for dr and df, we must attend only to the part involving dg, and must put this equal to $n\,dt$; the values first obtained for dx, dy, dz, dv, correspond to this assumption, and we have

$$\begin{aligned} \frac{dr}{dt} &= \frac{nae\sin f}{\sqrt{1-e^2}}, \\ \frac{df}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}, \\ \frac{dz}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}, \\ \frac{dx}{dt} &= \cos\phi\sec^2 y \, \frac{na^2\sqrt{1-e^2}}{r^2}, \\ \frac{dv}{dt} &= \cos\phi\sec^2 y \, \frac{na^2\sqrt{1-e^2}}{r^2}, \\ \frac{dy}{dt} &= \sin\phi\cos x \, \frac{na^2\sqrt{1-e^2}}{r^2}, \end{aligned}$$

and we then deduce

$$\begin{aligned}
\frac{d}{dt}\frac{dr}{dt} &= -\frac{na^3 e\cos f}{r^4},\\
\frac{d}{dt}\left(r^2\cos^2 y\,\frac{dv}{dt}\right) &= 0,\\
\frac{d}{dt}\left(r^2\frac{dy}{dt}\right) &= -\cos\phi\sin y\sec^2\phi\,\frac{n^2a^4(1-e^2)}{r^4},\\
\frac{d}{dt}\left(\frac{dy}{dt}\right) &= -\frac{2n^2a^3e\cos f}{r^4},
\end{aligned}$$

values which satisfy the undisturbed equations.

The disturbed equations may be dealt with in the usual manner by the method of the variation of the elements, and attending only to the variations of the elements we have

$$\begin{aligned}
dr &= 0,\\
dv &= 0,\\
dy &= 0,\\
d\frac{dr}{dt} &= \frac{d\Omega}{dr}\,dt,\\
d\left(r^2\cos^2 y\,\frac{dv}{dt}\right) &= \frac{d\Omega}{dv}\,dt,\\
d\left(r^2\frac{dy}{dt}\right) &= \frac{d\Omega}{dy}\,dt,
\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
dr &= 0,\\
dv &= 0,\\
dy &= 0,\\
d\,\frac{nae\sin f}{\sqrt{1-e^2}} &= \frac{d\Omega}{dr}\,dt,\\
d\,na^2\sqrt{1-e^2}\,\cos\phi &= \frac{d\Omega}{dv}\,dt,\\
d\,na^2\sqrt{1-e^2}\,\sin\phi\cos x &= \frac{d\Omega}{dy}\,dt,
\end{aligned}$$

where as before $\Omega = \Omega(r, v, y)$.

In virtue of the relations $dv = 0$, $dy = 0$, we have the above-mentioned equations,

$$\begin{aligned}
dx &= -\tan z\,\mathrm{cosec}\,\phi\,d\phi,\\
dz &= -\tan z\cot\phi\,d\phi,\\
d\mathfrak{C} &= \cos\phi\,dx,\\
dx &= d\theta,\\
dz + \cos\phi\,d\theta &= 0,
\end{aligned}$$

we have

$$\begin{aligned}
d\sin\phi\cos x &= \sin\phi\sin x\,dx + \cos x\cos\phi\,d\phi\\
&= \cos x\cos\phi\sec^2 z\,d\phi\\
&= \sec x\cos^2\phi\sec y\,d\phi;
\end{aligned}$$

and the last two equations for the variations become

$$\begin{aligned}
d\,na^2\sqrt{1-e^2}\,\cos\phi - na^2\sqrt{1-e^2}\,\sin\phi\,d\phi &= \frac{d\Omega}{dv}\,dt,\\
d\,na^2\sqrt{1-e^2}\,\sin\phi\cos x + na^2\sqrt{1-e^2}\,\sec x\cos^2\phi\sec^2 y\,d\phi &= \frac{d\Omega}{dy}\,dt,
\end{aligned}$$

and attending to the equations $\cos^2\phi + \sec^2 y \sin^2\phi \cos^2 x = 1$ we deduce at once

$$\begin{aligned} d\,na^2\sqrt{1-e^2} &= \cos\phi \sec^2 y\,\frac{d\Omega}{dv}\,dt + \sin\phi\cos x\,\frac{d\Omega}{dy}\,dt,\\ d\phi &= \frac{1}{na^2\sqrt{1-e^2}}\left(-\sin\phi\cos^2 x\,\frac{d\Omega}{dv}\,dt + \cos\phi\cos x\,\frac{d\Omega}{dy}\,dt\right). \end{aligned}$$

Now the position of the planet may be determined by the quantities r, z, θ, ϕ, or we may consider Ω as a function of the last-mentioned quantities. And if on the right-hand side $\Omega = \Omega(r, v, y)$ as before, the formulæ of transformation are

$$\frac{d\Omega}{dr}\,dr + \frac{d\Omega}{dz}\,dz + \frac{d\Omega}{d\theta}\,d\theta + \frac{d\Omega}{d\phi}\,d\phi = \frac{d\Omega}{dr}\,dr + \frac{d\Omega}{dv}\,dv + \frac{d\Omega}{dy}\,dy,$$

where

$$\begin{aligned} dv &= \cos\phi\sec^2 y\,dz - \tan z\cos^2 x\sin\phi\,d\phi + d\theta,\\ dy &= \sin\phi\cos x\,dz + (\tan z\cos x\cos\phi)\,d\phi, \end{aligned}$$

and we have

$$\begin{aligned} \frac{d\Omega}{d\theta} &= \frac{d\Omega}{dv},\\ \frac{d\Omega}{d\phi} &= \tan z\left(-\sin\phi\cos^2 x\,\frac{d\Omega}{dv} + \cos\phi\cos x\,\frac{d\Omega}{dy}\right),\\ \frac{d\Omega}{dz} &= \cos\phi\sec^2 y\,\frac{d\Omega}{dv} + \sin\phi\cos x\,\frac{d\Omega}{dy}, \end{aligned}$$

where on the left-hand side $\Omega = \Omega(r, z, \theta, \phi)$; and these equations give

$$\cot z\,\frac{d\Omega}{d\phi} = \cot\phi\,\frac{d\Omega}{dz} - \operatorname{cosec}\phi\,\frac{d\Omega}{d\theta},$$

an equation which is satisfied by $\Omega = \Omega(r, z, \theta, \phi)$. We have thus

$$\begin{aligned} dr &= 0,\\ dv &= 0,\\ dy &= 0,\\ d\,\frac{nae\sin f}{\sqrt{1-e^2}} &= \frac{d\Omega}{dr}\,dt,\\ d\,na^2\sqrt{1-e^2} &= \frac{d\Omega}{dz}\,dt,\\ d\phi &= -\frac{\cot z}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\phi}\,dt, \end{aligned}$$

which may be replaced by

$$\begin{aligned} dr &= 0,\\ d\theta &= \frac{\operatorname{cosec}\phi}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\phi}\,dt,\\ dz &= -\frac{\cot\phi}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\phi}\,dt,\\ d\,\frac{nae\sin f}{\sqrt{1-e^2}} &= \frac{d\Omega}{dr}\,dt,\\ d\,na^2\sqrt{1-e^2} &= \frac{d\Omega}{dz}\,dt,\\ d\phi &= -\frac{\cot z}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\phi}\,dt, \end{aligned}$$

where as before $\Omega = \Omega(r, z, \theta, \phi)$.

I remark that in the case of any central force whatever, we have an element h corresponding to $na^2\sqrt{1-e^2}$ in the elliptic theory, and the system for the variations is

$$\begin{aligned}
dr &= 0,\\
d\theta &= \frac{\operatorname{cosec}\phi}{h}\frac{d\Omega}{d\phi}\,dt,\\
dz &= -\frac{\cot\phi}{h}\frac{d\Omega}{d\phi}\,dt,\\
d\frac{dr}{dt} &= \frac{d\Omega}{dr}\,dt,\\
dh &= \frac{d\Omega}{dz}\,dt,\\
d\phi &= -\frac{\cot z}{h}\frac{d\Omega}{d\phi}\,dt,
\end{aligned}$$

where $\Omega = \Omega(r, z, \theta, \phi)$.

Imagine a point in the orbit, which I call the departure-point, the angular distances from this point are termed departures. And I write

$\mathfrak{p}$, the departure of planet,
ϖ, the departure of pericentre,
σ, the departure of node,

so that we have

$$\mathfrak{p} = \varpi + f, \qquad z = \mathfrak{p} - \sigma, \qquad \mathfrak{C} = \varpi - \sigma.$$

I write also

$\mathfrak{s}$, the longitude in orbit of departure-point, or, as it may be termed, the adjustment;

[Figure: a small spherical-triangle diagram with vertex angle θ and side σ; the departure-point is shown along the orbit line, with σ as the arc from node to departure-point.]

that is

$$\mathfrak{s} = \theta - \sigma.$$

In the undisturbed motion the departure-point is simply a fixed point in the orbit, but when the orbit is variable, the departure-point is taken to be the point of intersection of the orbit with any orthogonal trajectory of the successive positions of the orbit, a definition which is expressed analytically by the equation

$$d\sigma = \cos\phi\, d\theta.$$

The equation $z = \mathfrak{p} - \sigma$ gives

$$dz = d\mathfrak{p} - d\sigma = d\mathfrak{p} - \cos\phi\, d\theta,$$

or, what is the same thing,

$$dz = d\mathfrak{p} - \cos\phi\, d\theta.$$

But we have $dz + \cos\phi\, d\theta = 0$, and consequently $d\mathfrak{p} = 0$, an equation which expresses that the increment of departure, in so far as such increment arises from the variation of the elements, is equal to zero. Or, what is the same thing, the total increment of departure is equal to the infinitesimal angle between two consecutive radius vectors of the planet.

I propose to consider the departure-point as a point which is constantly defined as above, viz., when the orbit is variable, the departure-point is the point of intersection of the orbit with any orthogonal trajectory of the successive positions of the orbit; and as a particular case of the definition, when the orbit is fixed, the departure-point is simply a fixed point on the orbit. The orbit here considered is that of the planet and the position of the planet is determined by the departure and radius vector (the latitude being zero), and this is assumed to be the case whenever the departure is spoken of, and it is such departure which is denoted by the letter $\mathfrak{p}$. But we might consider a departure-point (defined as above), upon any other orbit whatever, and use such departure-point as an origin of longitude (for instance, in the lunar theory we might consider a longitude measured along the variable plane of the ...)

[Paper 212 continues beyond PDF p. 300.]